Alkane

61. Knocking sound occurs in engine when fuel
(a) Ignites slowly
(b) Ignites rapidly
(c) Contains water
(d) Is mixed with machine oil

62. Petroleum is mainly a mixture of
(a) Alkanes
(b) Cyclohexane
(c) Benzenoid hydrocarbons
(d) Alkenes

63. Which of the following has maximum boiling point
(a) *iso*-octane
(b) *n*-octane
(c) 2, 2, 3, 3-tetramethyl butane
(d) *n*-butane

64. Aqueous solution of the following compound on electrolysis gives ethane
(a) Acetic acid
(b) Acetamide
(c) Potassium acetate
(d) Ethyl acetate

65. Which of the following does not decolourise bromine solution in carbon disulphide
(a) Acetylene (b) Propene
(c) Ethane (d) Propyne

66. Anhydrous sodium acetate on heating with sodalime gives
(a) Acetic acid (b) Methane
(c) Calcium acetate (d) Ethane

67. Water gas is
(a) $CO + CO_2$
(b) $CO + N_2$
(c) $CO + H_2$
(d) $CO + N_2 + H_2$

68. A sample of gasoline contains 81% *iso*-octane and 19% *n*-heptane. Its octane number will be
(a) 19 (b) 81
(c) 100 (d) 62

69. The natural petroleum contains
(a) Saturated hydrocarbons
(b) Cyclic saturated hydrocarbons
(c) Compounds of sulphur
(d) All of these

70. The preparation of ethane by electrolysis of aqueous solution of potassium acetate is called as
(a) Wurtz reaction
(b) Sabatier-Senderen's reaction
(c) Kolbe's synthesis
(d) Grignard reaction

71. Action of hydrogen chloride on $CH_3-\underset{\underset{CH_3}{|}}{C}=CH_2$ and on $CH\equiv CH$ will predominantly give the compounds, respectively
(a) $CH_3-\underset{\underset{CH_3}{|}}{CH}=CH_2Cl$ and CH_2Cl-CH_2Cl
(b) $CH_3-\underset{\underset{CH_3}{|}}{CCl}=CH_3$ and CH_3-CHCl_2
(c) $CH_3-\underset{\underset{CH_3}{|}}{CH}=CH_2Cl$ and CH_3-CHCl_2
(d) $CH_3-\underset{\underset{CH_3}{|}}{CH}=CH_3$ and CH_2Cl-CH_2Cl

72. As the number of carbon atoms in a chain increases the boiling point of alkanes
(a) Increases
(b) Decreases
(c) Remains same
(d) May increase or decrease

73. In the fractional distillation of crude petroleum
(a) Petrol condenses at the bottom of the column
(b) The gases condense at the top of the column
(c) High boiling constituents condense at the bottom of the column
(d) High boiling constituents condense at the top of the column

74. Which of the following is not an endothermic reaction
(a) Dehydrogenation
(b) Ethane to ethene
(c) Combustion of propane
(d) Change of chlorine molecule into chlorine atoms.

75. Gasoline is the name of
(a) Crude oil
(b) The gaseous constituents of petroleum
(c) The mixture of uncondensed gases produced in the distillation of crude oil
(d)The mixture of the residue and gas oil obtained in the distillation of crude oil

76. In the process of cracking
(a)Organic compounds decompose into their constituent elements
(b)Hydrocarbons decompose into carbon and hydrogen
(c)High molecular weight organic compounds decompose to give low molecular weight organic compounds
(d) Hydrocarbons yield alkyl radicals and hydrogen

77. Octane number has 0 value for
(a) iso-octane
(b) *n*-hexane
(c) *n*-heptane
(d) iso-heptane

78. Dry distillation of sodium propanoate with sodalime gives
(a) Propane (b) Propene
(c) Ethane (d) Ethene

79. What is the chief product obtained when *n*-butane is treated with bromine in the presence of light at $130^{o}C$
(a) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-Br$
(b) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-Br$
(c) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-Br$
(d) $CH_3-CH_2-CH_2-CH_2-Br$

80. A mixture of propene and methane is obtained by the cracking of
(a) 1-butene (b) 2-butene
(c) *n*-butane (d) Isobutane

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